Market and Patent Research Report

Topic: Autonomous Robotic Systems for the Removal of Invasive Plant Species
Case Study: Giant Hogweed and Similar Neophytes
Course: Product Development – From Need to Market
Institution: Hochschule Kaiserslautern
Semester: SS 2025

1. Introduction

Invasive alien plant species such as *Heracleum mantegazzianum*, *Lupinus polyphyllus*, and *Jacobaea vulgaris* pose significant environmental and public health challenges in Europe. These plants negatively affect native biodiversity, outcompete local flora, and may pose toxicological risks to humans and animals. Conventional removal methods rely on herbicides, which are increasingly restricted by EU environmental legislation (e.g., Regulation (EU) No. 1143/2014), or labor-intensive mechanical clearing.

Recent advances in agricultural robotics provide an opportunity to address this issue using autonomous systems that employ machine learning, vision technologies, and non-chemical removal techniques, including electric pulse applications.

2. Market Overview and Technological Landscape

2.1. RootWave (UK) – Electric Weed Control System

- **Technology:** Applies high-frequency electric pulses through contact probes to thermally destroy plant cells from root to tip.
- **Target Species:** Effective on deep-rooted perennials including *Heracleum mantegazzianum*.
- **Advantages:**
    - Non-chemical and environmentally safe.
    - Minimal soil disturbance.
    - Suitable for robotic integration.Carbon Robotics
- **Use Cases:** Municipalities, ecological reserves, organic agriculture.

- **Video Demonstration:**

  [RootWave Demonstration1]

  [RootWave Demonstration2]

2.2. Carbon Robotics – LaserWeeder

- **Technology:** AI-powered machine vision with mounted CO_2 lasers for precision weed eradication.
- **Limitations:** Not designed specifically for tall or toxic plants like hogweed; more suitable for row crops.

- **Video Demonstration:** [LaserWeeder Overview]

2.3. Odd.Bot – Weed Pulling Robot (NL)

- **Technology:** Selectively detects and pulls weeds using robotic grippers, based on image classification algorithms.
- **Suitability:** Applicable for broadleaf weed species like *Jacobaea vulgaris*.

2.4. Drone and Aerial Detection Platforms (DK, DE, UK)

- **Application:** Drones combined with NDVI, hyperspectral imaging, or machine learning can map infestations of hogweed in open fields and forests.
- **Complementary Role:** Ideal for pre-mapping and targeting robotic ground intervention.

3. Patent Research Overview

3.1. EP3804483A1 – Tool for Eradication of Invasive Hogweed Species

- **Assignee:** Søren Lauritzen (Denmark)
- **Description:** Manual stem-puncturing tool to induce necrosis in giant hogweed without full severing. Minimizes regrowth by damaging vascular tissues.
- **Significance:** Basis for potential robotic adaptation using automatic puncturing probes.
- **Link:** Patent EP3804483A1

3.2. WO2021055485A1 – Autonomous Laser Weed Eradication System

- **Assignee:** Carbon RoboticsDreamstime+3Log in or sign up to view+3GeekWire+3
- **Description:** Robotic platform with sensors and vision system guiding a laser source to ablate unwanted plants.
- **Limitation:** Not optimized for toxic plants or non-row field applications.
- **Link:** WO2021055485A1

3.3. US20170238460A1 – Weeding Robot and Method

- **Assignee:** Blue River Technology
- **Description:** Selective weeding robot using capacitive and optical sensors for precision removal using mechanical tools.
- **Potential Use:** Adaptable platform for invasive species differentiation and removal.
- **Link:** US20170238460A1

3.4. EP3744173A1 – Weed Inactivation Device Using Electric Pulses

- **Assignee:** Zasso GmbH (Germany)
- **Description:** Electrically driven weed inactivation device using contact electrodes for plant-specific destruction.
- **Relevance:** Can be integrated into autonomous platforms for field use against large invasive targets.
- **Link:** [EP3744173A1](https://patents.google.com/patent/EP3744173A1/en